



Institute of Technology University of Moratuwa

Community-Based Ocean & Coastal Cleanup Management System

Software Requirements Specification

Version 1.0

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1 Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a detailed overview of the Community-Based Ocean & Coastal Cleanup Management System, describing its functional and non-functional requirements. This document is intended for:

- Development Team: To understand the complete system requirements for implementation.
- Project Supervisors: To evaluate the scope and completeness of the proposed solution.
- Stakeholders: Including government authorities (MEPA, Coastal Conservation Department), NGOs, volunteers, divers, and community members to validate that the system meets their needs [6].
- Quality Assurance Team: To derive test cases and validation criteria.

This SRS serves as the foundation for the design, development, and testing phases of the project.

1.2 Product Scope

1.2.1 Aim and Objectives

Aim:

To develop a Community-Based Ocean & Coastal Cleanup Management System that connects citizens, volunteers, divers, government authorities, and organizations for the identification, verification, and management of coastal pollution cleanup activities in Sri Lanka.

Objectives:

Objective ID	Description
OBJ-01	Gather requirements from all stakeholder groups (community members, volunteers, divers, government officers, organizations)
OBJ-02	Build reporting and verification module with photos, videos, GPS, community voting, trust percentage calculation, and 75% threshold-based verification
OBJ-03	Create government approval workflow for authority officers to review, approve/reject cleanup requests with comments
OBJ-04	Implement location-based alerts to detect nearby users, send notifications with map locations, and escalate alerts
OBJ-05	Build project monitoring with before/during/after images, completion percentages, and cleanup results recording
OBJ-06	Create diver opportunity module to connect divers with NGOs, tourism operators, and marine institutions
OBJ-07	Develop analytics dashboard for reporting on polluted sites, verified incidents, volunteer participation, and pollution trends
OBJ-08	Ensure security and scalability through access controls, data validation, and multi-tier architecture

Table 1: Description of Objectives

1.2.2 Project Boundary

In-Scope:

- User registration and profile management (with role-based access)
- Pollution reporting with photos, videos, GPS coordinates
- Community verification with trust percentage (75% threshold)
- Admin review and moderation
- Government body coordination and approval workflow
- Location-based alerts and notifications (with escalation)

- Project progress monitoring (before/during/after images)
- Diver employment and opportunity module
- Reporting and analytics dashboard
- Web-based system accessible via mobile browsers

Out of Scope:

- Physical cleanup operations (coordinated but not executed by system)
- Emergency response to hazardous material situations
- Integration with international environmental monitoring systems
- Native mobile applications (iOS/Android) - future scope
- Real-time GPS tracking of volunteers during cleanup
- Financial transactions between organizations and divers
- Legal enforcement features

The system boundaries are defined as follows:

- Geographic Boundary: Initially limited to Sri Lankan coastal areas (1,340 km coastline) [4]
- User Boundary: Community members, volunteers, divers, government officers (MEPA), administrators, and organizations (NGOs, tourism, marine institutions)
- Technical Boundary: Web-based application accessible via standard web browsers (Chrome, Firefox, Safari, Edge) on desktop and mobile devices
- Data Boundary: Pollution cleanup requests, user profiles, cleanup projects, verification records, approval documents, and analytics data
- Integration Boundary: External APIs including geocoding services and map APIs (Google Maps or equivalent)

2 Overall Description

2.1 Product Perspective

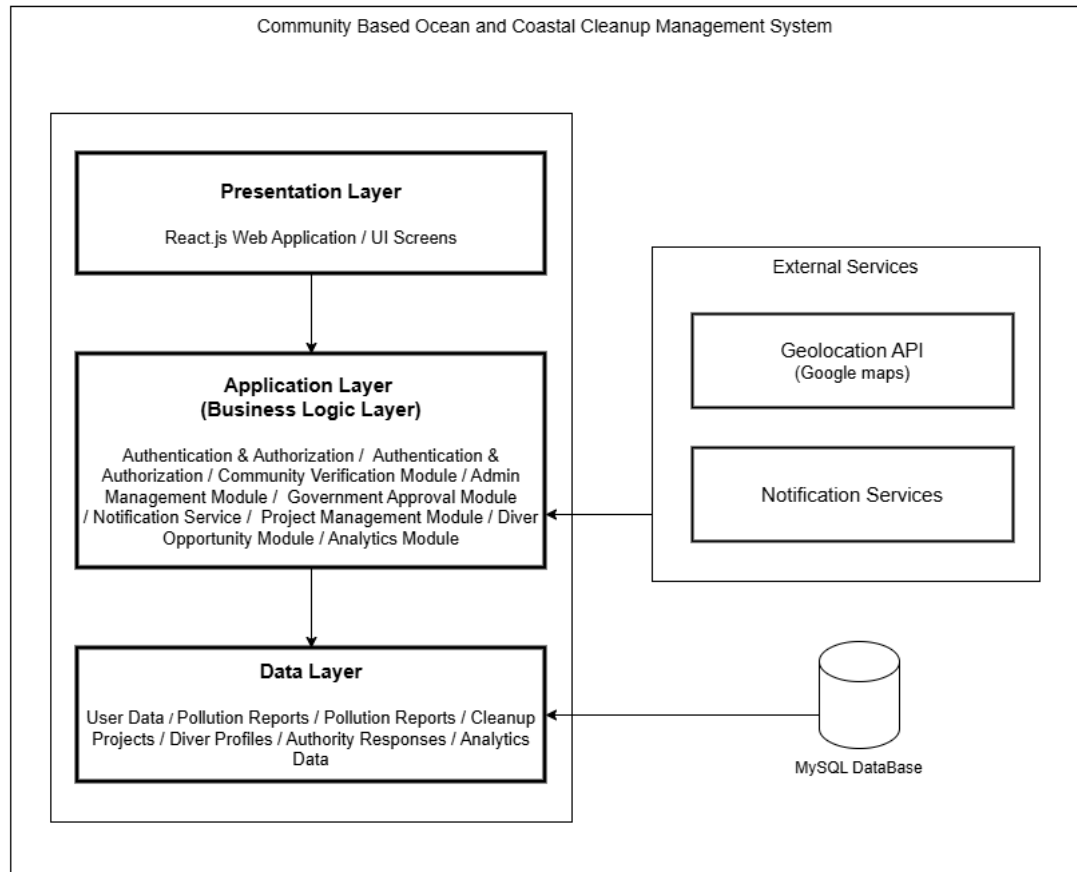


Figure 1: High-Level Architectural Diagram of System

The system follows a multi-tier architecture consisting of a Presentation Layer, Application Layer, and Data Layer. The Presentation Layer is developed using a React.js web application that provides user interfaces for all system users. The Application Layer is built using Spring Boot and handles the main business logic, including authentication, pollution reporting, community verification, admin management, government approval, notifications, project management, diver opportunities, and analytics. The Data Layer uses a MySQL database to store all system data such as user details, reports, cleanup projects, diver profiles, authority responses, and analytics information. The system also integrates external services

such as a Geolocation API for map-based features and a Notification Service for alerts and updates, where all external communication is managed through the Application Layer [2].

2.2 User Classes and Characteristics

The Community-Based Ocean & Coastal Cleanup Management System will be used by different user groups with different responsibilities and access levels. The main user classes are volunteer non-divers, volunteer divers, government authority officers, system administrators, and organizations. Each user group has different technical knowledge, system access, and functional needs.

- Volunteer non-divers:

Volunteer non-divers are registered citizens who use the system to report polluted ocean and coastal locations. They may have only basic computer or mobile phone knowledge; therefore, the system should provide a simple and user-friendly interface. These users can submit Pollution cleanup requests by uploading photographs, videos, descriptions, and location details. They can also participate in community verification by voting, commenting, and confirming whether a reported pollution incident is genuine. The system should support easy navigation and simple language so that users from different regions of Sri Lanka can use it effectively.

- Non-Registered Users:

Non-registered users are public users who access the system without creating an account. Their system access is limited compared to registered users. They may be allowed to view general information, awareness content, public cleanup updates, and selected verified Pollution cleanup requests. They cannot submit Pollution cleanup requests, vote, comment, or join cleanup projects unless they register with the system.

This user class helps increase public awareness while protecting the system from misuse.

- Volunteer Divers:

Volunteer divers are users with diving skills who are willing to participate in ocean or underwater cleanup activities. These users may have basic to moderate digital literacy and should be able to manage their profile, certification details, diving experience, equipment availability, and preferred working regions. They can receive location-based alerts about cleanup projects and participate in suitable activities. Since diving activities may involve safety risks, the system should clearly display project details, location information, and safety-related notices before participation.

- Government Authority Officers:

Government authority officers are professional users from relevant environmental or coastal management bodies such as MEPA or other authorized institutions. They use the system to review cleanup requests or requests that require official attention, approval, or guidance. These users are expected to have moderate computer literacy and strong knowledge of environmental procedures. The system should provide them with structured report details, evidence, location information, verification status, and options to approve, reject, or provide official comments.

- System Administrator:

The system administrator is the authorized user responsible for managing and controlling the overall system. This user should have good technical knowledge and administrative experience. The administrator can manage user accounts, review reports, moderate suspicious or false content, handle complaints, manage abusive users, and monitor overall system activities. The administrator also ensures that the system remains reliable, secure, and properly maintained.

- Organizations:

Organizations include NGOs, tourism operators, marine institutions, environmental groups, and research teams involved in coastal cleanup or marine conservation activities. These users may use the system to create or support cleanup projects, coordinate volunteers and divers, post diver-related opportunities, and view reports or analytics. Organization users are expected to have moderate digital literacy and basic project coordination knowledge. The system should allow them to access relevant project details, communicate with participants, and monitor cleanup progress.

2.3 Operating Environment

The Community-Based Ocean & Coastal Cleanup Management System will operate as a web-based application. Users will access the system through standard web browsers using desktop computers, laptops, tablets, or smartphones. Since the system is web-based, users do not need to install any special software on their devices.

The client-side interface will run on modern web browsers such as Google Chrome, Microsoft Edge, and Safari. The system will support both desktop and mobile web browsers to allow users to submit reports, view updates, and receive notifications conveniently.

The backend of the system will operate on a server environment using Spring Boot. The frontend will be developed using React.js, and the system data will be stored in a MySQL database[15][14]. The system may be deployed on a cloud platform such as AWS, Microsoft Azure, or Google Cloud to support remote access, scalability, and availability.

The system will also interact with external software services such as map or geolocation APIs to display pollution locations and support location-based features. Communication between the frontend, backend, database, and external services will be handled through secure network connections using HTTP/HTTPS protocols [9][10].

The system is expected to operate in an environment with stable internet connectivity, supported web browsers, and access to required external services. No special hardware is required other than standard internet-enabled devices.

2.4 Design and Implementation Constraints

The Community-Based Ocean & Coastal Cleanup Management System must be designed and implemented within several technical, operational, and security limitations. These constraints help define the boundaries of the system and guide the development process.

The system will be developed as a web-based application. Therefore, users must have an internet-enabled device and a modern web browser to access the system. The system will support desktop and mobile browsers, but a separate native mobile application is not included in the current scope.

The system must follow a multi-tier architecture. The presentation layer, application layer, data layer, and external service layer should be clearly separated. This design will improve maintainability, scalability, and reliability.

The selected technologies must be used consistently during implementation. The frontend will be developed using React.js, the backend will be developed using Spring Boot, and the database will be managed using MySQL. React is suitable for component-based user interfaces, Spring supports REST-based web services, and MySQL is used as the database platform [15][11].

The system will depend on external services such as map or geolocation APIs. These services are required for displaying pollution locations, selecting map coordinates, and supporting location-based alerts. If these external services are unavailable, some location-based functions may be affected.

Security must be considered throughout the design and development process. The system should include user authentication, role-based access control, secure password handling, and protected communication using HTTPS. HTTPS provides encrypted communication between the client and server [9].

The system should be simple and easy to use because some users may have only basic digital literacy. The interface should use clear language and simple navigation so that community users, volunteers, divers, organizations, and government officers can use the system effectively.

2.5 Assumptions and Dependencies

The Community-Based Ocean & Coastal Cleanup Management System is developed based on several assumptions and dependencies. These factors may affect the system requirements, design, and implementation if they change during the project.

It is assumed that users will provide accurate information when submitting Pollution cleanup requests. This includes correct location details, clear descriptions, and valid images or videos. The accuracy of user-submitted data is important for report verification, cleanup planning, and authority review.

It is assumed that users will have access to internet-enabled devices such as smartphones, tablets, laptops, or desktop computers. Since the system is web-based, stable internet connectivity and a modern web browser are required for proper use.

The system depends on external map or geolocation services to display pollution locations and support location-based alerts [8][2]. For example, map APIs usually require an API key and proper setup before they can be used in a web application. If these services are unavailable or restricted, location-related features may be affected.

It is assumed that government officers, organizations, volunteers, and divers will be willing to use the system for reporting, verification, coordination, and monitoring activities. If stakeholder participation is low, the effectiveness of community verification and cleanup coordination may be reduced.

The system depends on the selected software technologies, including React.js for the frontend, Spring Boot for the backend, and MySQL for data storage. The backend application will communicate with the database to store and retrieve system data. Spring documentation also supports the use of Spring applications with MySQL databases.

It is assumed that required system resources, such as hosting services, database storage, and backup facilities, will be available during development and deployment. Any limitation in server capacity or storage may affect system performance.

It is also assumed that the system will follow the defined project scope. The system will support reporting, verification, coordination, monitoring, and analytics.

3 External Interface Requirements

3.1 User Interfaces

The Community-Based Ocean & Coastal Cleanup Management System will provide a web-based graphical user interface for all users. The interface should be simple, clear, and easy to use because different types of users, including community members, volunteers, divers, organizations, government officers, and administrators, will use the system (*Appendix A*).

The user interface will be designed to support both desktop and mobile web browsers. The layout should be responsive so that users can access the system from computers, tablets, and smartphones. Common interface elements should be used consistently throughout the system. The system should provide clear labels, simple instructions, and meaningful error messages

The following main user interfaces are required for the system:

Interface	Description	Main Users
Login and Registration Interface	Allows users to create accounts and log in securely.	All registered users
User Dashboard	Displays user activities, cleanup requests, alerts, and project updates.	Community users, volunteers, divers
Pollution cleanup request Submission Interface	Allows users to submit Pollution cleanup requests with images, videos, descriptions, and map locations.	Volunteer non-divers
Community Verification Interface	Allows users to vote, comment, and verify Pollution cleanup requests.	Registered users
Admin Dashboard	Allows the administrator to review reports, manage users, and moderate system activities.	System administrator

Government Review Interface	Allows government officers to review escalated reports and provide approval, rejection, or comments.	Government authority officers
Project Monitoring Interface	Displays cleanup progress, before/during/after images, status updates, and completion percentage.	Users, organizations, admin
Analytics Dashboard	Displays statistics such as reported sites, verified incidents, volunteer participation, and regional pollution trends.	Admin, organizations, authorities

Table 2: User Interface Requirements

3.2 Hardware Interfaces

The System does not require any special hardware device. Since the system is a web-based application, it can be accessed using standard internet-enabled devices such as desktop computers, laptops, tablets, and smartphones.

Users may use the built-in camera or file storage of their device to upload photos and videos related to Pollution cleanup requests. Mobile devices may also use location services to help identify the pollution location. Browser-based geolocation features normally require user permission and a secure connection before location data is shared with the web application [8].

The system will not directly control any physical hardware. All hardware interaction will happen through the user's web browser and device capabilities. Image uploads will be handled through the browser's file upload function, and location selection will be supported through map or geolocation services. HTML file inputs can also support image or video selection from device storage or camera where the browser and device allow it [7].

3.3 Software Interfaces

The System will interact with several software components to support user interface functions, business logic, data storage, location services, and communication between system layers.

The frontend of the system will be developed using React.js. React.js will be used to create the web-based user interfaces such as login, registration, Pollution cleanup requesting, dashboards, verification pages, and project monitoring pages. React supports component-based user interface development, which helps create reusable and maintainable interface elements [14].

The backend of the system will be developed using Spring Boot. The backend will handle business logic, user authentication, report processing, verification calculations, notification handling, admin review, government coordination, and data access. Spring Boot is suitable for developing web applications and REST-based services.

The system will use MySQL as the database management system. MySQL will store user details, Pollution cleanup requests, uploaded evidence details, votes, comments, cleanup project data, diver profiles, organization details, authority feedback, and analytics data. MySQL documentation provides official database reference material and connector support for application integration [11].

The system will also integrate with a map or geolocation API, such as Google Maps API, to display pollution locations, select map coordinates, and support location-based alerts. The Maps JavaScript API supports interactive maps, markers, and location-based display features for web applications [2].

Communication between the frontend and backend will be handled through REST APIs using HTTP/HTTPS. The backend will exchange data with the frontend in JSON format. The backend will communicate with the MySQL database using database queries through suitable database connectors.

Software Component	Purpose	Data Exchanged	Communication Method
React.js Frontend	Provides web-based user interfaces	User inputs, form data, dashboard data, report details	REST API using HTTP/HTTPS
Spring Boot Backend	Handles business logic and system processing	User requests, report data, verification data, project data	REST API and database connection
MySQL Database	Stores system data permanently	User records, reports, votes, comments, project details, analytics data	SQL queries through backend
Map/Geolocation API	Displays locations and supports map-based features	Coordinates, map markers, location data	External API calls
Notification Service	Sends system alerts and updates	Alert messages, user contact details, project updates	API-based or email-based communication
Web Browser	Allows users to access the system	User actions, uploaded files, displayed responses	HTTP/HTTPS

Table 3: Software Interfaces Requirements

The system should ensure that all software interfaces communicate securely and reliably. Sensitive data such as login details, user profiles, report information, and authority feedback should be protected during transmission and storage. External services should be properly configured before deployment to avoid failures in map display, location selection, or notification features.

3.4 Communications Interfaces

The System will use standard web communication methods to exchange data between users, the web application, the backend server, the database, and external services. Since the system is web-based, most communication will happen through the internet using web protocols.

The main communication between the user's web browser and the system server will be handled through HTTP/HTTPS. HTTP is commonly used for communication between web browsers and web servers, while HTTPS provides encrypted communication using TLS, making it more suitable for protecting sensitive data such as login details and user information.

The frontend and backend will communicate through REST APIs. The data exchanged between these layers will mainly use JSON format because it is lightweight, readable, and commonly used in web applications.

The system may also use SMTP if email notifications are included [3]. SMTP is the standard protocol used for internet email transport. It can be used to send account verification messages, system alerts, approval updates, or other important notifications to users.

For location-based functions, the backend or frontend may communicate with external map or geolocation services through API requests. These requests will be used to display maps, select coordinates, validate locations, and support location-based alerts.

All communication involving sensitive data should be protected using secure protocols. User passwords, personal information, report details, uploaded evidence, and authority comments should not be transmitted through insecure channels. HTTPS should be used for all client-server communication.

The system should maintain reliable and secure communication between all components. If a communication failure occurs, such as a lost internet connection or unavailable external service, the system should display a clear error message to the user.

4.1 Use case Diagram

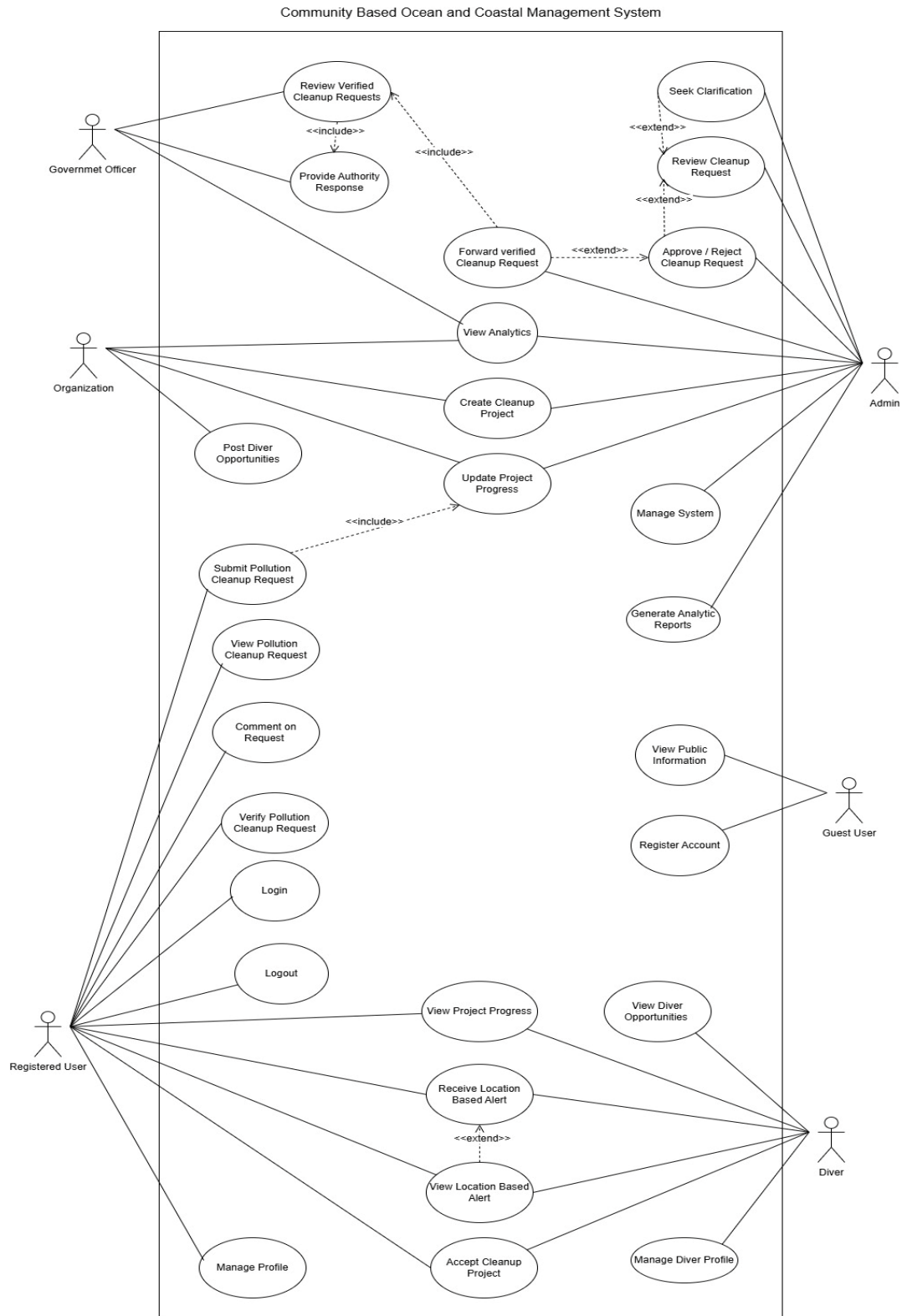


Figure 2: Use case Diagram for the System

4.1.1 Use case Description

In this section each use case of the system explains how different users interact with the system to perform required functions (*Appendix B*).

Use Case ID	Use Case	Short Description
UC-01	View Public Information	A non-registered user visits the system to view public information, awareness content, and selected cleanup updates.
UC-02	Register Account	A new user creates an account by entering required details and selecting the relevant user role.
UC-03	Login	A registered user enters valid credentials to access the system according to their role.
UC-04	Logout	A logged-in user exits the system securely by ending the current session.
UC-05	Manage Profile	A registered user updates personal information such as contact details, location, and availability.
UC-06	Manage Diver Profile	A diver updates diving-related details such as certification, experience, equipment availability, and preferred regions.
UC-07	Submit Pollution Cleanup Request	A community user submits a pollution cleanup request by adding a description, evidence, date, time, and location details.
UC-08	View Pollution Cleanup Request	A registered user views submitted pollution cleanup request with evidence, location, status, and related details.
UC-09	Verify Pollution Cleanup Request	A registered user votes whether a pollution cleanup request is true or false based on the available evidence.

UC-10	Comment on Cleanup Request	A registered user adds comments to a Pollution cleanup request to provide additional information or clarification.
UC-11	Review Verified Cleanup Request	The administrator reviews submitted, verified, unverified, or suspicious Pollution cleanup requests.
UC-12	Approve / Reject Cleanup Request	The administrator approves or rejects a Pollution cleanup request after reviewing the evidence and verification results.
UC-13	Seek Clarification	The administrator requests additional information from the report submitter when details are incomplete or unclear.
UC-14	Manage System	The administrator manages user accounts, handles abusive users, and controls access where necessary.
UC-15	Forward Verified Approval Request	The administrator forwards an approved cleanup request to the relevant government authority for official review.
UC-16	Review Verified Cleanup Request	A government officer reviews the forwarded cleanup request, including evidence, location, and admin remarks.
UC-17	Provide Authority Response	A government officer provides approval, rejection, comments, or official instructions for a cleanup request.
UC-18	Receive Location-Based Alert	Users or divers receive notifications about nearby verified Pollution cleanup requests or cleanup projects.
UC-19	View Location-Based Alert	A user or diver view alert and may choose to viewing more details about the cleanup activity.
UC-20	Create Cleanup Project	An organization or administrator creates a cleanup project after the report is approved and required permission is available.

UC-21	Accept Cleanup Project	A user or diver joins an available cleanup project as a participant.
UC-22	Update Project Progress	An organization or administrator updates project progress with status, notes, images, and completion percentage.
UC-23	View Project Progress	Registered users, organizations, or administrators view cleanup progress, updates, and completion status.
UC-24	Post Diver Opportunity	An organization posts an opportunity for divers related to cleanup or marine conservation activities.
UC-25	View Diver Opportunities	A diver views available opportunities posted by organizations.
UC-26	View Analytics	Authorized users view statistics related to reports, verified incidents, projects, participation, and regional trends.
UC-27	Generate Analytic Reports	Authorized users generate reports based on pollution incidents, cleanup activities, participation, and analytics data.

Table 4: Use case of the System

5 System Features

5.1 User Registration and Profile Management

5.1.1 Description and Priority

This feature allows users to register with the system, log in, and manage their profile details. Users can register under suitable roles such as volunteer non-diver, diver, organization, government officer, or administrator. This feature has high priority because only registered users can access the main system functions.

5.1.2 Stimulus and Response Sequence

The registration option will be available on the main page of the system. When a user selects the registration option, the system will display a registration form. The user must enter required details such as name, email, password, contact details, and user role. After submission, the system validates the details and creates a user account. Registered users can log in and update their profile details. Divers can add additional information such as certification level, experience, equipment availability, and preferred working regions.

5.1.3 Functional Requirements

REQ-1: The system shall allow users to register with a suitable user role.

REQ-2: The system shall allow registered users to log in using valid credentials.

REQ-3: The system shall validate user registration details before creating an account.

REQ-4: The system shall support role-based access control.

REQ-5: The system shall allow users to update their profile information.

REQ-6: The system shall allow divers to enter certification, experience, equipment, and preferred region details.

REQ-7: The system shall display an error message for invalid login or registration details.

5.2 Pollution Reporting and Incident Submission

5.2.1 Description and Priority

This feature allows registered community users to submit Pollution cleanup requests related to ocean and coastal areas. Users can provide descriptions, images, videos, date, time, and location details. This feature has high priority because Pollution cleanup requesting is the starting point of the cleanup process.

5.2.2 Stimulus and Response Sequence

When a registered user notices a polluted coastal or ocean location, the user can open the Pollution reporting page. The system will display a cleanup request submission form. The user enters the pollution description, uploads images or videos, and selects the location using a map or location input. After the user submits the form, the system validates the details and stores the cleanup request for community verification.

5.2.3 Functional Requirements

REQ-8: The system shall allow registered users to submit Pollution cleanup requests.

REQ-9: The system shall allow users to upload photos and videos as evidence.

REQ-10: The system shall allow users to enter a description of the pollution incident.

REQ-11: The system shall allow users to select or capture the pollution location.

REQ-12: The system shall record the date and time of request submission.

REQ-13: The system shall prevent cleanup request submission if required fields are missing.

REQ-14: The system shall store submitted request for verification.

5.3 Community Verification and Trust Validation

5.3.1 Description and Priority

This feature allows registered users to verify Pollution cleanup requests submitted by other users. Users can vote whether a report is true or false and can add comments.

The system calculates a trust percentage based on the votes. This feature has high priority because it helps reduce false reports and improve report reliability.

5.3.2 Stimulus and Response Sequence

After a Pollution cleanup request is submitted, it will be visible for community verification. Registered users can open the cleanup request, view the evidence and location, and vote whether that request is true or false. Users may also add comments to support their decision. The system records the votes and updates the trust percentage. If the request reaches the required trust level, it will be marked as community verified. If not, it will be forwarded for admin review.

5.3.3 Functional Requirements

REQ-15: The system shall allow registered users to view Pollution cleanup requests for verification.

REQ-16: The system shall allow users to vote true or false on a Pollution cleanup request.

REQ-17: The system shall prevent duplicate votes from the same user on the same cleanup request.

REQ-18: The system shall calculate the trust percentage of each cleanup request.

REQ-19: The system shall mark requests as community verified when the required trust threshold is reached.

REQ-20: The system shall allow users to comment on Pollution cleanup requests.

REQ-21: The system shall forward unverified or suspicious reports to the administrator for review.

5.4 Admin Review and Moderation

5.4.1 Description and Priority

This feature allows the system administrator to review cleanup requests, manage users, and moderate system activities. The administrator can approve, reject, or seek

clarification for Pollution cleanup requests. This feature has high priority because it supports system reliability and prevents misuse.

5.4.2 Stimulus and Response Sequence

The administrator logs in and opens the admin dashboard. The system displays submitted, verified, unverified, or suspicious requests. The administrator reviews the cleanup request details, evidence, votes, comments, and location. Based on the review, the administrator can approve the request, reject or seek clarification from the request submitter. The administrator can also manage users who submit false cleanup requests or behave abusively.

5.4.3 Functional Requirements

REQ-22: The system shall allow the administrator to view submitted requests.

REQ-23: The system shall allow the administrator to review evidence, votes, comments, and location details.

REQ-24: The system shall allow the administrator to approve or reject cleanup requests.

REQ-25: The system shall allow the administrator to seek clarification from request submitters.

REQ-26: The system shall allow the administrator to manage abusive users.

REQ-27: The system shall allow the administrator to restrict or remove false cleanup request.

REQ-28: The system shall maintain records of administrator decisions.

5.5 Government Coordination and Approval Management

5.5.1 Description and Priority

This feature allows approved cleanup requests to be forwarded to relevant government authorities for permission, review, or official guidance [5][6].

Government officers can review the request and provide approval, rejection, or instructions. This feature has high priority because cleanup activities in coastal and marine areas may require official permission or authority guidance [5].

5.5.2 Stimulus and Response Sequence

After a Pollution cleanup request is reviewed and approved by the administrator, the administrator can forward the cleanup approval request to the relevant government authority. The government officer logs in and views the pending approval request. The system displays the request details, evidence, location, community verification result, and admin remarks. The officer reviews the request and provides approval, rejection, or official instructions. The system records the response and notifies the administrator.

5.5.3 Functional Requirements

REQ-29: The system shall allow the administrator to forward approved cleanup requests to government authority officers.

REQ-30: The system shall allow government officers to view forwarded cleanup approval requests.

REQ-31: The system shall display request details, evidence, location, verification status, and admin remarks to the government officer.

REQ-32: The system shall allow government officers to approve or reject cleanup approval requests.

REQ-33: The system shall allow government officers to provide comments or official instructions.

REQ-34: The system shall notify the administrator about the authority response.

REQ-35: The system shall maintain the approval status of each cleanup request.

5.6 Alert and Notification Management

5.6.1 Description and Priority

This feature sends alerts to users and divers near a verified pollution location or cleanup project. It helps inform nearby users and encourages participation in cleanup activities. This feature has high priority because effective communication is important for volunteer coordination.

5.6.2 Stimulus and Response Sequence

When a cleanup request is approved and a cleanup project is ready, the system identifies nearby registered users and divers based on their location details. The system sends notifications with project information and map location. If there is no sufficient response from nearby users, the system may expand the alert area. Users can open the notification and respond by accepting or declining participation.

5.6.3 Functional Requirements

REQ-36: The system shall identify registered users and divers near a cleanup location.

REQ-37: The system shall send location-based alerts to nearby users.

REQ-38: The system shall include cleanup details and map location in the alert.

REQ-39: The system shall allow users and divers to respond to alerts.

REQ-40: The system shall record user responses to cleanup alerts.

REQ-41: The system shall expand the alert area if there is insufficient response.

REQ-42: The system shall send project update notifications to relevant users.

5.7 Project Progress Monitoring

5.7.1 Description and Priority

This feature allows authorized users to monitor and update the progress of cleanup projects. It includes progress notes, images, status updates, and completion

percentage. This feature has medium priority because it supports transparency and accountability after a cleanup project starts.

5.7.2 Stimulus and Response Sequence

When a cleanup project is created, the organization or administrator can open the project monitoring page. The system displays the project details and current progress. The authorized user can upload before, during, and after images, add progress notes, and update the completion percentage. The system saves the progress update and displays it to relevant users.

5.7.3 Functional Requirements

REQ-43: The system shall allow authorized users to update cleanup project progress.

REQ-44: The system shall allow users to upload before, during, and after cleanup images.

REQ-45: The system shall allow users to add progress notes.

REQ-46: The system shall allow users to update the project completion percentage.

REQ-47: The system shall display project status and progress timeline.

REQ-48: The system shall record cleanup results after project completion.

5.8 Diver Opportunity Management

5.8.1 Description and Priority

This feature allows organizations to post opportunities for divers and allows divers to view suitable opportunities. Divers can use their profiles to show skills, certification, and previous participation. This feature has medium priority because it supports diver engagement and creates additional value for the system.

5.8.2 Stimulus and Response Sequence

An organization logs in and opens the diver opportunity section. The organization enters opportunity details such as title, description, location, required skills, and date. The system validates and publishes the opportunity. Divers can view the opportunity

and check whether it matches their skills and preferred regions. The system may notify suitable divers about new opportunities.

5.8.3 Functional Requirements

REQ-49: The system shall allow organizations to post diver opportunities.

REQ-50: The system shall allow divers to view available opportunities.

REQ-51: The system shall allow organizations to enter opportunity details.

REQ-52: The system shall display diver profiles to authorized organizations.

REQ-53: The system shall record divers' previous cleanup participation.

5.9 Reporting and Analytics

5.9.1 Description and Priority

This feature provides reports and analytics related to pollution incidents, verified reports, cleanup projects, volunteer participation, and regional pollution trends. This feature has medium priority because it supports decision-making for administrators, organizations, and government officers.

5.9.2 Stimulus and Response Sequence

An authorized user opens the analytics dashboard. The system displays summary statistics such as the number of reported pollution sites, verified incidents, active projects, completed projects, and volunteer participation. The user can apply filters based on region, time period, or report category. The system displays the filtered results and may generate reports for review.

5.9.3 Functional Requirements

REQ-54: The system shall display the number of reported pollution sites.

REQ-55: The system shall display the number of verified pollution incidents.

REQ-56: The system shall display the number of active and completed cleanup projects.

REQ-57: The system shall display volunteer and diver participation statistics.

REQ-58: The system shall display region-wise pollution trends.

REQ-59: The system shall allow authorized users to filter analytics by region and time period.

REQ-60: The system shall allow authorized users to generate reports.

REQ-61: The system shall restrict analytics access to authorized users only.

6 Other Nonfunctional Requirements

6.1 Performance Requirements

The System should provide acceptable performance for all users during normal system usage. Since the system will be used for reporting pollution incidents, viewing requests, verifying information, receiving alerts, and monitoring cleanup projects, it should respond within a reasonable time.

The system should load main pages such as the login page, dashboard, report submission page, and project monitoring page within a short time under normal internet conditions. User actions such as login, report submission, voting, commenting, and profile updates should be processed without unnecessary delay.

The system should support multiple users accessing the platform at the same time. Community users, volunteers, divers, organizations, administrators, and government officers should be able to use the system without major performance issues during normal operating conditions.

Location-based features should also respond efficiently. When a verified Pollution cleanup request is created, the system should identify nearby users and send notifications within a reasonable time. Map loading and location selection should work smoothly when external map services are available.

Uploaded files such as images and videos should be handled efficiently. The system should allow users to upload evidence for Pollution cleanup requests and progress updates, but file size limits should be applied to avoid slow performance and excessive storage usage.

The system should also maintain stable performance when storing and retrieving data from the database. Reports, votes, comments, project updates, user profiles, and analytics data should be loaded accurately and within an acceptable time.

Performance Requirements:

REQ-NF-1: The system shall load commonly used pages within 3 to 5 seconds under normal internet conditions.

REQ-NF-2: The system shall process user login, registration, voting, and commenting actions within a reasonable response time.

REQ-NF-3: The system shall allow multiple users to access the system at the same time without major performance degradation.

REQ-NF-4: The system shall send location-based alerts to relevant users within a reasonable time after a report is verified.

REQ-NF-5: The system shall support image and video uploads with defined file size limits to maintain system performance.

REQ-NF-6: The system shall retrieve cleanup requests, user details, project updates, and analytics data efficiently from the database.

REQ-NF-7: The system shall display a proper message if a request takes too long or if a service is temporarily unavailable.

6.2 Safety Requirements

The System should support safe cleanup coordination and reduce possible risks to users, volunteers, divers, organizations, and government officers. Since ocean and coastal pollution may include unsafe materials such as sharp waste, oil, chemicals, or contaminated waste, the system must provide safety-related information before users participate in cleanup activities.

The system should clearly warn users that it is designed for planned cleanup coordination and not for emergency response. MEPA is responsible for preventing, controlling, reducing, and managing marine pollution in Sri Lanka, and it also operates response-related services such as oil and hazardous substance spill planning [6].

The system should display safety notices for cleanup projects, especially when volunteers or divers are involved. Users should be informed to follow authority instructions, use proper safety measures, and avoid participating in activities that may cause harm. The system should also allow administrators to mark cleanup request as unsafe or requiring authority review.

Safety Requirements:

REQ-NF-8: The system shall display safety warnings for cleanup activities that may involve hazardous materials.

REQ-NF-9: The system shall allow administrators to mark Pollution cleanup requests as hazardous or unsafe.

REQ-NF-10: The system shall forward hazardous incidents, such as oil spills or chemical pollution, to the relevant authority for review.

REQ-NF-11: The system shall prevent general users from organizing cleanup activities for reports marked as hazardous until authority guidance is provided.

REQ-NF-12: The system shall display clear safety instructions before users join cleanup projects.

REQ-NF-13: The system shall notify volunteers and divers if a cleanup project has special safety concerns.

REQ-NF-14: The system shall maintain records of authority comments and safety-related instructions.

REQ-NF-15: The system shall clearly state that it does not replace official emergency response services.

6.3 Security Requirements

The System must protect user accounts, personal data, Pollution cleanup requests, uploaded evidence, and authority feedback. Since the system stores information such as user profiles, contact details, location data, images, comments, and verification records, security and privacy must be considered throughout the design and implementation process.

The system should follow secure web application practices. The Data Protection Authority of Sri Lanka was established under the Personal Data Protection Act to regulate personal data processing and protect individual privacy [1], so the system

should handle personal data responsibly and only collect information needed for system functions.

User authentication must be implemented to ensure that only registered users can access protected functions. Different user roles such as community users, divers, organizations, government officers, and administrators must have different access permissions. For example, only administrators should be allowed to manage users, and only authorized government officers should review escalated reports.

Passwords must not be stored as plain text. They should be stored using secure password hashing methods, following accepted web application security practices. OWASP recommends protecting stored passwords even if the database is compromised [12].

All communication between the client and server should use HTTPS [13]. Secure REST services should provide HTTPS endpoints to protect authentication details, API keys, and other sensitive data during transmission.

The system should also include input validation to prevent invalid or harmful data from being submitted through forms, comments, uploads, and API requests. Uploaded files such as images and videos should be checked for allowed file types and file size limits. The system should maintain audit records for important actions such as report approval, rejection, authority comments, user restrictions, and admin moderation.

Security Requirements:

REQ-NF-16: The system shall require user authentication before allowing access to protected system functions.

REQ-NF-17: The system shall support role-based access control for community users, divers, organisations, government officers, and administrators.

REQ-NF-18: The system shall store passwords using secure hashing methods and shall not store passwords as plain text.

REQ-NF-19: The system shall use HTTPS for communication between the frontend, backend, and external services.

REQ-NF-20: The system shall validate user inputs before processing or storing data.

REQ-NF-21: The system shall restrict access to administrative functions only to authorized administrators.

REQ-NF-22: The system shall restrict access to government review functions only to authorized government officers.

REQ-NF-23: The system shall protect personal data such as user profiles, contact details, location data, uploaded evidence, and authority feedback.

REQ-NF-24: The system shall apply file type and file size validation for uploaded images and videos.

REQ-NF-25: The system shall maintain audit logs for important actions such as report approval, rejection, user restriction, and authority feedback.

REQ-NF-26: The system shall display clear error messages for unauthorized access attempts without revealing sensitive system details.

REQ-NF-27: The system shall allow users to log out securely from their accounts.

6.4 Other Non-Functional Requirements

REQ-NF-28: The system shall provide a user-friendly interface with clear labels, simple navigation, and understandable instructions.

REQ-NF-29: The system shall be reliable and should correctly store and retrieve user data, reports, votes, comments, project updates, and authority responses.

REQ-NF-30: The system shall be available to users through supported web browsers when the server and internet connection are available.

REQ-NF-31: The system shall be maintainable by separating the frontend, backend, database, and external services using a multi-tier architecture.

REQ-NF-32: The system shall be scalable to support an increasing number of users, requests, cleanup projects, and uploaded media.

REQ-NF-33: The system shall be portable across common devices such as desktops, laptops, tablets, and smartphones through responsive web design.

REQ-NF-34: The system shall be robust enough to handle invalid inputs, missing fields, upload errors, and unavailable external services without crashing.

REQ-NF-35: The system shall be testable by allowing each major module, such as request submission, verification, admin review, notifications, and analytics, to be tested separately.

REQ-NF-36: The system shall be interoperable with external services such as map APIs and notification services through standard web-based interfaces [10][2].

REQ-NF-37: The system shall support future improvements, such as mobile application development, additional authority dashboards, and extended analytics, without requiring major redesign.

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Appendix A: User Interface mock-ups

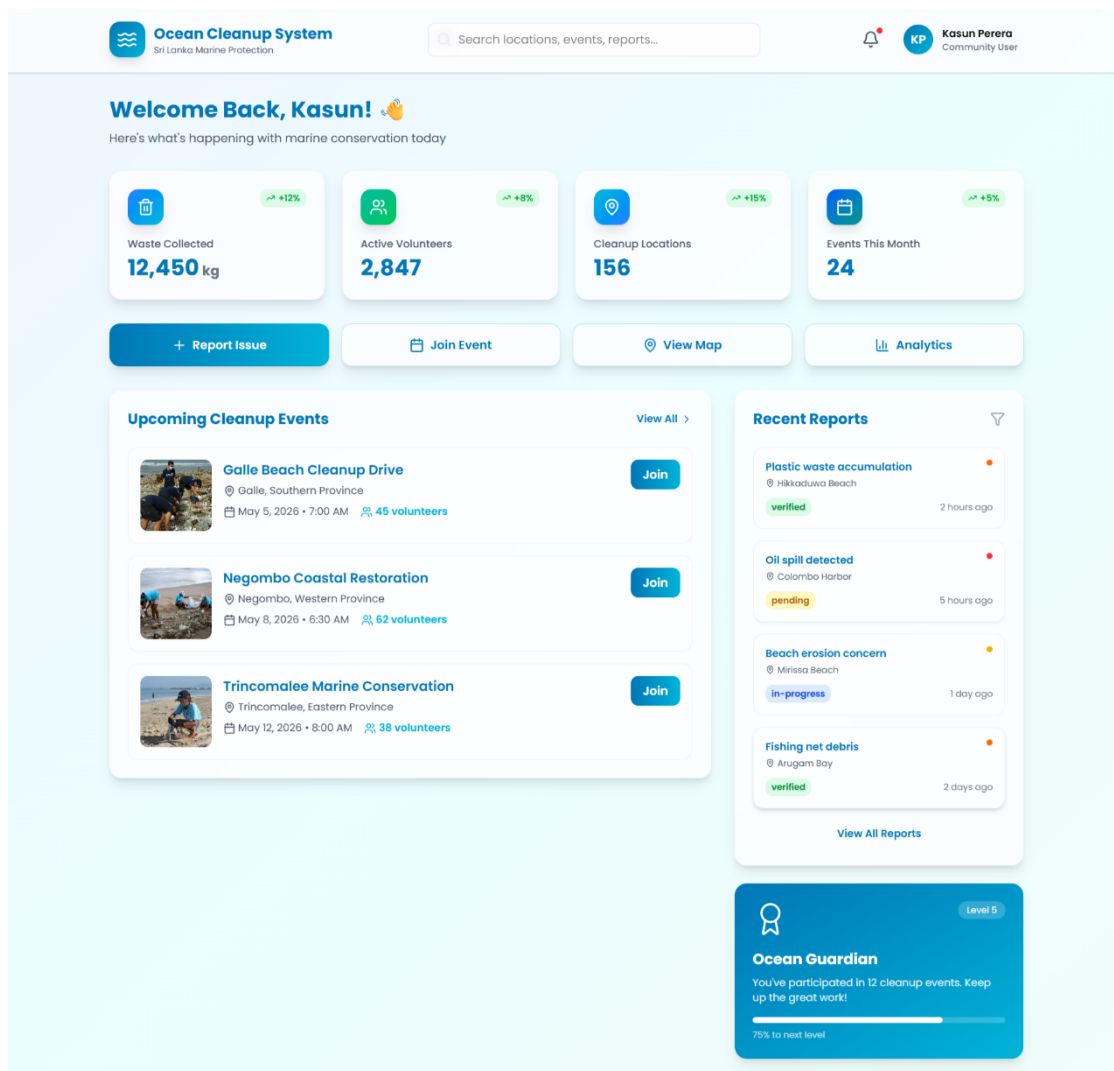


Figure 3: UI mock-up 1_Volunteer non-diver Dashboard


Ocean Cleanup System
Sri Lanka Marine Protection




[← Back to Dashboard](#)



Report Pollution Incident

Help protect our oceans by reporting pollution and coastal issues


 Your report will be reviewed by our verification team and government officers. Accurate information helps us respond faster.


Upload Photos & Videos



Drag and drop files here

or

Browse Files

Supports: JPG, PNG, MP4, MOV (Max 10MB per file)


Incident Details

Incident Title *

Description *

Minimum 50 characters

Pollution Type *

Severity Level *


Location Information

Location Name *

GPS Coordinates *

[Use Current Location](#)



Interactive map will display here

Enter coordinates to view location


Date & Time

Date Observed *

Time Observed *

Figure 4: UI mock-up 2_Pollution cleanup reporting Module

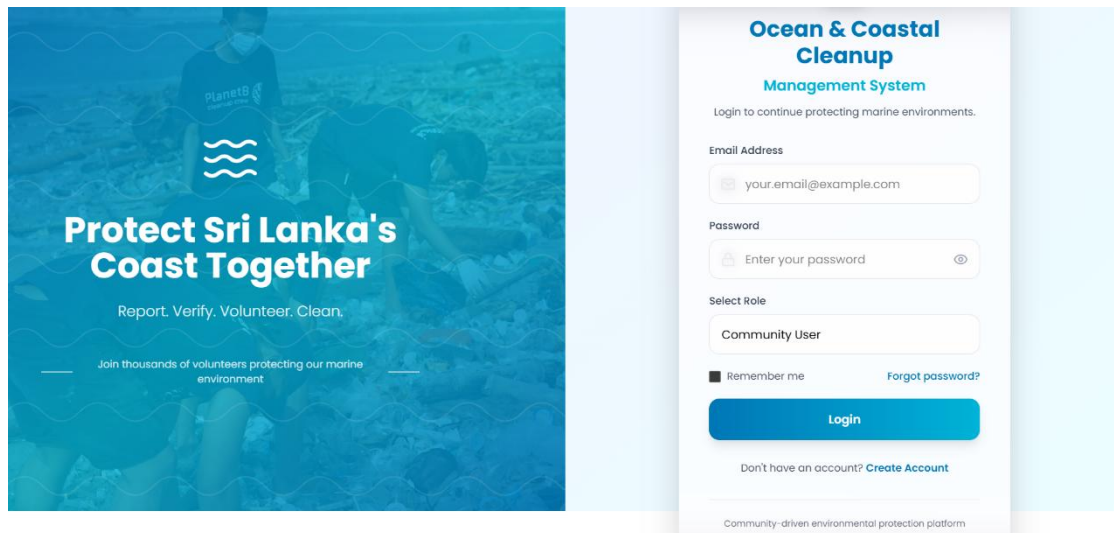


Figure 5: UI mock-up 3_Login Page

Appendix B: Use Case Descriptions

Use Case ID	UC-01	
Use Case	View Public Information	
Priority	Medium	
Primary System Actor	Non-Registered User	
Other participant Actors	None	
Description	The non-registered user views general public information about coastal cleanup activities, awareness content, and selected public updates.	
Pre-conditions	The user must have access to the web application.	
Trigger	The user opens the public website.	
Flow of Events	Actor Action	System Response
	1. User visits the system website.	System displays the public home page.
	2. User selects available public information.	System displays selected information.
	3. User views cleanup awareness content or public updates.	System loads the requested content.
Post-conditions	The user views public information without logging in.	
Alternative Flows	If the requested page is unavailable, the system displays an error message.	
Assumptions	Public information is available and maintained by the system administrator.	

Table 5: Use Case-01

Use Case ID	UC-02	
Use Case	Register Account	
Priority	High	
Primary System Actor	Volunteer non-diver / Diver / Organization / Government Officer	
Other participant Actors	System Administrator	
Description	A new user creates an account by providing required details and selecting a suitable user role.	
Pre-conditions	The user must not already have an account with the same email or username.	
Trigger	The user selects the registration option.	
Flow of Events	Actor Action	System Response
	1. User opens the registration page.	System displays the registration form.
	2. User enters required details.	System validates the entered information.
	3. User selects the user role.	System records the selected role.
	4. User submits the form.	System creates the account or displays validation errors.
Post-conditions	A user account is created successfully.	
Alternative Flows	If required fields are missing or invalid, the system displays an error message.	
Assumptions	Users provide accurate registration details.	

Table 6: Use Case-02

Use Case ID	UC-03	
Use Case	Login	
Priority	High	
Primary System Actor	Registered User	
Other participant Actors	None	
Description	A registered user logs in to access system functions based on their role.	
Pre-conditions	The user must have a registered account.	
Trigger	The user selects the login option.	
Flow of Events	Actor Action	System Response
	1. User enters username/email and password.	System validates the login credentials.
	2. User submits login form.	System authenticates the user.
	3. Valid credentials are provided.	System redirects user to the relevant dashboard.
Post-conditions	The user is logged in successfully.	
Alternative Flows	If credentials are incorrect, the system displays an error message.	
Assumptions	The user remembers valid login credentials.	

Table 7: Use Case-03

Use Case ID	UC-04	
Use Case	Logout	
Priority	Low	
Primary System Actor	Registered User	
Other participant Actors	None	
Description	A logged-in user exits the system securely.	
Pre-conditions	The user must be logged in.	
Trigger	The user selects the logout option.	
Flow of Events	Actor Action	System Response
	1. User clicks logout. 2. Session is closed.	System ends the user session. System redirects user to the login or home page.
Post-conditions	The user is logged out of the system.	
Alternative Flows	If the Session already expired, the system redirects the user to login.	
Assumptions	The system manages user sessions securely.	

Table 8: Use Case-04

Use Case ID	UC-05	
Use Case	Manage Profile	
Priority	Medium	
Primary System Actor	Registered User	
Other participant Actors	None	
Description	A registered user updates personal details such as contact information, location, and availability.	
Pre-conditions	The user must be logged in.	
Trigger	The user opens the profile page.	
Flow of Events	Actor Action	System Response
	1. User opens profile settings.	System displays current profile details.
	2. User edits profile information.	System validates the updated details.
	3. User saves changes.	System updates the profile.
Post-conditions	The user profile is updated successfully.	
Alternative Flows	If invalid data is entered, the system displays validation errors.	
Assumptions	Users provide correct profile details.	

Table 9: Use Case-05

Use Case ID	UC-06	
Use Case	Manage Diver Profile	
Priority	Medium	
Primary System Actor	Diver	
Other participant Actors	Organization	
Description	A diver updates diving-related details such as certification, experience, equipment availability, and preferred regions.	
Pre-conditions	The diver must be registered and logged in.	
Trigger	The diver opens the diver profile section.	
Flow of Events	Actor Action	System Response
	1.Diver opens diver profile page. 2.Diver enters certification and experience details. 3.Diver saves the profile.	System displays current diver details. System validates the details. System updates the diver profile.
Post-conditions	The diver profile is updated and available for relevant opportunities.	
Alternative Flows	If required diver details are missing, the system displays an error message.	
Assumptions	Divers provide accurate certification and experience details.	

Table 10: Use Case-06

Use Case ID	UC-07	
Use Case	Submit Pollution Cleanup Request	
Priority	High	
Primary System Actor	Volunteer non-diver	
Other participant Actors	None	
Description	A registered community user submits a pollution cleanup request with description, evidence, and location details.	
Pre-conditions	The user must be logged in.	
Trigger	The user selects “Submit Pollution cleanup request.”	
Flow of Events	Actor Action	System Response
	1. User opens request submission form.	System displays the request form.
	2. User enters pollution details.	System validates the required fields.
	3. User uploads images or videos.	System stores the evidence.
	4. User selects location.	System records location coordinates.
	5. User submits the request.	System saves the report for verification.
Post-conditions	The Pollution cleanup request is stored in the system.	
Alternative Flows	If required data is missing, the system prevents submission and displays an error message.	
Assumptions	User provides genuine pollution information.	

Table 11: Use Case-07

Use Case ID	UC-08	
Use Case	View Pollution cleanup requests	
Priority	High	
Primary System Actor	Registered User	
Other participant Actors	None	
Description	Registered users view Pollution cleanup requests submitted by the community.	
Pre-conditions	The user must be logged in.	
Trigger	The user selects the Pollution cleanup requests section.	
Flow of Events	Actor Action	System Response
	1. User opens the cleanup requests page. 2. User selects a request.	System displays available requests. System displays request details, evidence, location, and status.
Post-conditions	The user views Pollution cleanup request details.	
Alternative Flows	If no requests are available, the system displays a suitable message.	
Assumptions	Requests are available in the system database.	

Table 12: Use Case-08

Use Case ID	UC-09	
Use Case	Verify Pollution cleanup requests	
Priority	High	
Primary System Actor	Registered User	
Other participant Actors	System Administrator	
Description	A registered user votes whether a Pollution cleanup request appears true or false.	
Pre-conditions	User must be logged in and the request must be available for verification.	
Trigger	User selects a cleanup request for verification.	
Flow of Events	Actor Action	System Response
	1. User views request details. 2. User votes true or false. 3. 4. Vote is submitted.	System displays evidence and voting options. System records the vote. System updates the trust percentage.
Post-conditions	The cleanup request verification status is updated.	
Alternative Flows	If the user has already voted, the system prevents duplicate voting.	
Assumptions	Users vote honestly based on available evidence.	

Table 13: Use Case-09

Use Case ID	UC-10	
Use Case	Comment on Request	
Priority	Medium	
Primary System Actor	Registered User	
Other participant Actors	System Administrator	
Description	A registered user votes whether a Pollution cleanup request appears true or false.	
Pre-conditions	User must be logged in, and the cleanup request must be visible.	
Trigger	User selects the comment option.	
Flow of Events	Actor Action	System Response
	1. User opens a Pollution cleanup request.	System displays request details and comment section.
	2. User enters a comment.	System validates the comment.
	3. User submits the comment.	System saves and displays the comment.
Post-conditions	The comment is added to the cleanup request.	
Alternative Flows	If the comment contains inappropriate content, the system may flag it for admin review.	
Assumptions	Comments are used to support verification and discussion.	

Table 14: Use Case-10

Use Case ID	UC-11	
Use Case	Review Cleanup request	
Priority	High	
Primary System Actor	Admin	
Other participant Actors	Registered User	
Description	The administrator reviews submitted, verified, unverified, or suspicious Pollution cleanup requests.	
Pre-conditions	Admin must be logged in.	
Trigger	Admin opens the cleanup request review dashboard.	
Flow of Events	Actor Action	System Response
	1. Admin opens pending request.	System displays requests requiring review.
	2. Admin selects a request.	System displays evidence, votes, comments, and location.
	3. Admin examines the cleanup request.	System allows approval, rejection, or clarification.
Post-conditions	The cleanup request is ready for an admin decision.	
Alternative Flows	If request details are insufficient, admin may seek clarification.	
Assumptions	Admin reviews requests carefully before making decisions.	

Table 15: Use Case-11

Use Case ID	UC-12	
Use Case	Approve or Reject Cleanup Requests	
Priority	High	
Primary System Actor	Admin	
Other participant Actors	Registered User	
Description	The administrator approves or rejects Pollution cleanup requests after reviewing submitted information.	
Pre-conditions	The request must be reviewed by admin.	
Trigger	Admin selects approve or reject decision.	
Flow of Events	Actor Action	System Response
	1. Admin reviews request information. 2. Admin approves the request. 3. Admin rejects the request.	System displays decision options. System updates request status as approved. System updates request status as rejected and records reason.
Post-conditions	Request status is updated.	
Alternative Flows	Admin may seek clarification instead of approving or rejecting.	
Assumptions	Admin decision is based on request evidence and verification results.	

Table 16: Use Case-12

Use Case ID	UC-13	
Use Case	Seek Clarification	
Priority	Medium	
Primary System Actor	Admin	
Other participant Actors	Volunteer non-diver	
Description	The administrator requests additional information from the cleanup request submitter before making a decision.	
Pre-conditions	Cleanup request must be under admin review.	
Trigger	Admin finds that request details are incomplete or unclear.	
Flow of Events	Actor Action	System Response
	1. Admin selects Seek clarification.	System displays clarification message form.
	2. Admin enters required clarification.	System sends clarification request to submitter.
	3. Submitter provides additional details.	System updates the request.
Post-conditions	Additional information is added to the ret.	
Alternative Flows	If the submitter does not respond, admin may reject or keep the request pending.	
Assumptions	Submitter is willing to provide more details.	

Table 17: Use Case-13

Use Case ID	UC-14	
Use Case	Manage System	
Priority	High	
Primary System Actor	Admin	
Other participant Actors	Registered Users	
Description	The administrator manages user accounts, handles abusive users, and controls. system	
Pre-conditions	Admin must be logged in.	
Trigger	Admin opens user management section.	
Flow of Events	Actor Action	System Response
	1. Admin views user list.	System displays registered users.
	2. Admin selects a user account.	System displays user details and activity.
	3. Admin updates system settings.	System changes the features.
Post-conditions	User account status is updated.	
Alternative Flows	If no action is required, admin exits without changes.	
Assumptions	Admin manages users and system.	

Table 18: Use Case-14

Use Case ID	UC-15	
Use Case	Forward Cleanup Approval Request	
Priority	High	
Primary System Actor	Admin	
Other participant Actors	Government Officer	
Description	The admin forwards verified cleanup requests to government authority officer.	
Pre-conditions	The request must be verified and approved by admin.	
Trigger	Admin selects forward to authority.	
Flow of Events	Actor Action	System Response
	1. Admin selects approved request.	System displays forwarding option.
	2. Admin enters remarks if needed.	System attaches admin remarks.
	3. Admin forwards request.	System sends request to government officer.
Post-conditions	Cleanup approval request is sent to the authority.	
Alternative Flows	If required cleanup request information is missing, system prevents forwarding.	
Assumptions	Relevant authority account is available in the system.	

Table 19: Use Case-15

Use Case ID	UC-16	
Use Case	Review Cleanup Approval Request	
Priority	High	
Primary System Actor	Government Officer	
Other participant Actors	Admin	
Description	The government officer reviews a cleanup approval request	
Pre-conditions	The request must be verified and approved by admin.	
Trigger	Admin selects forward to authority.	
Flow of Events	Actor Action	System Response
	1. Officer opens approval dashboard.	System displays pending requests.
	2. Officer selects a request.	System displays request details, evidence, location, and admin remarks.
	3. Officer reviews the request.	System allows officer to provide a decision or instructions.
Post-conditions	The request is reviewed by the officer.	
Alternative Flows	Officer may request more information before giving a final decision.	
Assumptions	Government officer has authority to review the request.	

Table 20: Use Case-16

Use Case ID	UC-17	
Use Case	Provide Authority Response	
Priority	High	
Primary System Actor	Government Officer	
Other participant Actors	Admin	
Description	The government officer provides approval, rejection, comments, or official instructions for a cleanup request.	
Pre-conditions	Officer must review the cleanup approval request.	
Trigger	Government officer selects the pending approval request.	
Flow of Events	Actor Action	System Response
	1. Officer enters decision or comments.	System validates response details.
	2. Officer submits response.	System records authority response.
	3. Response is saved.	System notifies admin.
Post-conditions	Authority response is recorded in the system.	
Alternative Flows	If response details are incomplete, system displays an error message.	
Assumptions	Officer provides accurate and official feedback.	

Table 21: Use Case-17

Use Case ID	UC-18	
Use Case	Receive Location-Based Alert	
Priority	High	
Primary System Actor	User / Diver	
Other participant Actors	None	
Description	Users and divers receive alerts about nearby verified Pollution cleanup requests or cleanup projects.	
Pre-conditions	User must be registered and location details must be available.	
Trigger	A request is approved or a cleanup project is created.	
Flow of Events	Actor Action	System Response
	1. System identifies nearby users.	System prepares alert message.
	2. Notification service sends alert.	User receives notification with location details.
	3. User opens alert.	System displays related cleanup project details.
Post-conditions	User receives cleanup or pollution alert.	
Alternative Flows	If no nearby users are found, system may expand the alert area.	
Assumptions	User location data is available and notification service works.	

Table 22: Use Case-18

Use Case ID	UC-19	
Use Case	View Location-Based Alert	
Priority	Medium	
Primary System Actor	User / Diver	
Other participant Actors	Admin / Organization	
Description	A user or diver view a received cleanup alert.	
Pre-conditions	User must receive an alert.	
Trigger	User opens the alert notification. .	
Flow of Events	Actor Action	System Response
	1. User opens alert. 2. User may select view more details.	System displays project details. System displays a detailed description about request including location date, time and images.
Post-conditions	User response is recorded.	
Alternative Flows	User may decline or ignore the alert.	
Assumptions	User response is voluntary.	

Table 23: Use Case-19

Use Case ID	UC-20	
Use Case	Create Cleanup Project	
Priority	High	
Primary System Actor	Organization / Admin	
Other participant Actors	Government Officer	
Description	An organization or admin creates a cleanup project based on approved reports and authority guidance.	
Pre-conditions	A Pollution cleanup request must be approved and required permission should be available.	
Trigger	Organization or admin selects create project.	
Flow of Events	Actor Action	System Response
	1. Actor opens project creation page.	System displays project form.
	2. Actor enters project details.	System validates the details.
	3. Actor submits project.	System creates cleanup project.
Post-conditions	Cleanup project is created in the system.	
Alternative Flows	If required permission is missing, system may prevent project creation.	
Assumptions	Project details are accurate and approved.	

Table 24: Use Case-20

Use Case ID	UC-21	
Use Case	Accept Cleanup Project	
Priority	Medium	
Primary System Actor	User / Diver	
Other participant Actors	Organization / Admin	
Description	A user or diver joins a cleanup project as a participant.	
Pre-conditions	User must be logged in and project must be open for participation.	
Trigger	User selects join project.	
Flow of Events	Actor Action	System Response
	1. User views project details.	System displays project information and safety notes.
	2. User selects accept option.	System records participation request.
	3. Participation is accepted.	System adds user to project participant list.
Post-conditions	User is added as a project participant.	
Alternative Flows	If project is full or closed, system prevents joining.	
Assumptions	User agrees to follow project instructions.	

Table 25: Use Case-21

Use Case ID	UC-22	
Use Case	Update Project Progress	
Priority	Medium	
Primary System Actor	Organization / Admin	
Other participant Actors	Project Participants	
Description	Organization or admin updates cleanup project progress with status, notes, and completion percentage.	
Pre-conditions	Cleanup project must already exist.	
Trigger	Actor selects update project progress.	
Flow of Events	Actor Action	System Response
	1. Actor opens project progress page.	System displays current progress.
	2. Actor enters new progress details.	System validates the update.
	3. Actor submits update.	System saves updated progress.
Post-conditions	Project progress is updated.	
Alternative Flows	If invalid progress percentage is entered, system displays an error message.	
Assumptions	Updates are entered by authorized users.	

Table 26: Use Case-22

Use Case ID	UC-23	
Use Case	View Project Progress	
Priority	Medium	
Primary System Actor	Registered User / Organization / Admin	
Other participant Actors	None	
Description	Users view cleanup project progress, including updates, images, and completion status.	
Pre-conditions	Project progress data must be available.	
Trigger	User opens project progress section.	
Flow of Events	Actor Action	System Response
	1. User selects a cleanup project.	System displays project details.
	2. User opens progress section.	System displays progress updates and images.
	3. User views completion status.	System displays completion percentage and timeline.
Post-conditions	User views project progress information.	
Alternative Flows	If no progress updates are available, system displays a suitable message.	
Assumptions	Project updates are maintained by authorized users.	

Table 27: Use Case-23

Use Case ID	UC-24	
Use Case	Post Diver Opportunity	
Priority	Medium	
Primary System Actor	Organization	
Other participant Actors	Diver	
Description	An organization posts an opportunity for divers related to cleanup, marine conservation, or related activities.	
Pre-conditions	Organization must be registered and logged in.	
Trigger	Organization selects post opportunity.	
Flow of Events	Actor Action	System Response
	1. Organization opens opportunity form.	System displays required fields.
	2. Organization enters opportunity details.	System validates the details.
	3. Organization submits the post.	System publishes the opportunity.
Post-conditions	Diver opportunity is posted in the system.	
Alternative Flows	If required fields are missing, system displays validation errors.	
Assumptions	Organization provides genuine opportunity details.	

Table 28: Use Case-24

Use Case ID	UC-25	
Use Case	View Diver Opportunity	
Priority	Medium	
Primary System Actor	Diver	
Other participant Actors	Organization	
Description	A diver views available opportunities posted by organizations.	
Pre-conditions	Diver must be logged in.	
Trigger	Diver opens opportunity section.	
Flow of Events	Actor Action	System Response
	1. Diver opens opportunities page.	System displays available opportunities.
	2. Diver selects an opportunity.	System displays full details.
	3. Diver reviews opportunity information.	System allows diver to contact or express interest if supported.
Post-conditions	Diver views opportunity details.	
Alternative Flows	If no opportunities are available, system displays a suitable message.	
Assumptions	Opportunities are posted by authorized organizations.	

Table 29: Use Case-25

Use Case ID	UC-26	
Use Case	View Analytics	
Priority	Medium	
Primary System Actor	Admin / Organization / Government Officer	
Other participant Actors	None	
Description	Authorized users view analytics related to Pollution cleanup requests, verified incidents, projects, participation, and regional trends.	
Pre-conditions	Actor must be logged in and authorized.	
Trigger	Actor opens analytics dashboard.	
Flow of Events	Actor Action	System Response
	1. Actor opens analytics dashboard.	System displays summary statistics.
	2. Actor selects category or filter.	System displays filtered analytics.
	3. Actor views graphs or tables.	System presents relevant data.
Post-conditions	Actor views system analytics.	
Alternative Flows	If no data is available, system displays an empty data message.	
Assumptions	Analytics and project data are available in the database.	

Table 30: Use Case-26

Use Case ID	UC-27	
Use Case	Generate analytic Reports	
Priority	Medium	
Primary System Actor	Admin	
Other participant Actors	None	
Description	Admin generate reports based on pollution incidents, cleanup projects, participation, and regional trends.	
Pre-conditions	Admin must be logged in.	
Trigger	Admin selects generate report option.	
Flow of Events	Actor Action	System Response
	1. Admin opens report generation section.	System displays report filters.
	2. Admin selects report type and time period.	System retrieves relevant data.
	3. Admin generates reports.	System displays or exports report output.
Post-conditions	A report is generated successfully.	
Alternative Flows	If no matching data is available, system displays an empty message.	
Assumptions	System contains sufficient data for analytics.	

Table 31: Use Case-27